

Table of Laplace Transforms

We can choose a *laplace transform* to a given function based on the following table. Assume a and c are constants.

$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1	$\frac{1}{s}$
$e^{a t}$	$\frac{1}{s - a}$
$t^n, n = 1, 2, \dots$	$\frac{n!}{s^{n+1}}$
$\sin(a t)$	$\frac{a}{s^2 + a^2}$
$\cos(a t)$	$\frac{s}{s^2 + a^2}$
$\sinh(a t)$	$\frac{a}{s^2 - a^2}$
$\cosh(a t)$	$\frac{s}{s^2 - a^2}$
heavyside function $\rightarrow u_c(t) = u(t - c)$	$\frac{e^{-c s}}{s}$
dirac delta function $\rightarrow \delta(t - c)$	$e^{-c s}$
1st derivative $\rightarrow f'(t)$	$sF(s) - f(0)$
2nd derivative $\rightarrow f''(t)$	$s^2F(s) - sf(0) - f'(0)$

Note: This list only contains some of the more commonly used laplace transforms and formulas.